



ROMINA SALJOOGHIAN

AI Engineer · Agentic AI · Robotics & Control

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portfolio

PROFILE

I am an AI Engineer with an MSc in Automation and Control Engineering. I build LLM and agentic AI systems, computer vision pipelines, and production REST APIs, and I like taking a project from a first idea all the way to something that runs and is properly tested. The work I enjoy most is where AI meets the physical world, like my language-to-action robotic agent that turns natural language into safe robot actions. Coming from control engineering, I pay close attention to how a system behaves and what it does when something goes wrong, and I bring that same rigor to everything I build.

SKILLS & CERTIFICATIONS

Programming	Python (<i>freeCodeCamp</i>) C++ MATLAB SQL (<i>freeCodeCamp</i>)
ML & Deep Learning	PyTorch TensorFlow Keras scikit-learn OpenCV Hugging Face
LLM & Agentic AI	LangGraph & LangSmith (<i>LangChain Academy</i>) MCP RAG OpenAI API Anthropic API
Backend & API	FastAPI REST API Pydantic pytest
MLOps & Tools	Docker MLflow Hydra Git GitHub Actions Linux $\LaTeX$ Automation Explorer (<i>UiPath</i>)
Control & Robotics	ROS2 (Humble) PyBullet Simulink MPC LQR Kalman Filter PID System ID

EDUCATION

MSc · Automation and Control Engineering

Politecnico di Milano

Sep 2022 – Mar 2026

Milan, Italy

- **GPA** 107 / 110 — **Thesis:** *Semantic Characterization of Robotic Environments*
- Open-vocabulary scene graph generation for robotic manipulation using foundation models (CLIP, SAM, Grounding DINO, RAM) — MERLIN Lab.
- **Selective Coursework:** Neural Networks & Deep Learning · Control of Industrial Robots · Multiagent Systems · Software Engineering · Model Identification · Data-Driven Control · Autonomous Vehicles

BSc · Electrical Engineering (Minor: Control)

K. N. Toosi University of Technology

Sep 2016 – Aug 2020

Tehran, Iran

- **GPA** 3.65 / 4.0 — **Thesis:** *Classification of Brain Regions in the Mouse via Neural Network*
- **Selective Coursework:** Control Systems · Digital Signal Processing · Linear Algebra · Probability & Statistics · PLC Lab

WORK EXPERIENCE

Delphi Software Developer

Torfeh Negar Holding

Jan 2020 – Sep 2021

Tehran, Iran

- Built production UI and backend modules for Holoo accounting software serving thousands of users
- Optimized SQL Server queries; resolved bottlenecks and stabilized nightly batch processes

Instrumentation & Control Intern

Tavan Gostare Zino

Jun 2019 – Sep 2019

Tehran, Iran

- Designed and tested industrial grounding and earthing systems for IEC standards compliance

PROJECTS & RESEARCH

lang2action — Language-to-Action Robotic Agent — Python · C++ · LangGraph · MCP · PyBullet · ROS2

2026

github.com/Romi-s/lang2action

- Vision-Language-Action agent: a custom **MCP server** exposes scene-graph tools, a **LangGraph** planner grounds commands with a **hallucination guard**, a swappable action layer executes pick-and-place in PyBullet
- Eval-first: **0.95 grounding, 0.95 task success, 1.00 refusal**; closed the sim-to-real perception gap **from 12% to 72% recall without training** (ablation-verified)
- **ROS2 (Humble) + C++** action-server controller (rclcpp) with physical goal verification; 47 tests, Docker, MIT licensed

Scene Graph Generation for Robotic Manipulation — Python · PyTorch · CLIP · YOLO-World · TensorRT · ROS2

2025 – 2026

Live Demo

- MSc thesis (MERLIN Lab): a modular open-vocabulary pipeline chaining 4 foundation models (RAM, Grounding DINO, SAM, CLIP) with a Transformer pair-scorer that turns an RGB image into a scene graph for LLM task planners; rebuilt a collapsed end-to-end baseline modular for a **6× gain** (PredCls mR@100 48.9%), with **zero-shot transfer** to manipulation scenes on a VLM-annotated benchmark
- Re-engineered it into a deployable, real-time service: one small detector plus geometry and a monocular-depth cue replaces the four-model GPU stack, compiled ONNX to **TensorRT** (INT8/FP16) for a **4.7× speedup**, and shipped as a **ROS2** perception node at 3.3 Hz with 29 tests, CI, and a live Gradio demo

Compliance RAG Agent — Python · FastAPI · LangGraph · ChromaDB · BM25 · LangSmith  Compliance-RAG-Agent  Live Demo	2026
- Full-stack RAG service answering regulatory questions with source citations: a 4-node LangGraph pipeline over hybrid retrieval (ChromaDB vectors + BM25, merged by Reciprocal Rank Fusion), deployed on Google Cloud Run with keyless (OIDC) CI/CD - Instrumented it for production LLMOps: LangSmith tracing and Prometheus metrics, plus an evaluation suite (LLM-as-judge faithfulness and a retrieval-recall metric) gating a CI regression check, which caught a real recall gap (about 0.79) the system handled by refusing rather than hallucinating	
Agentic Job-Application Automation System — PowerShell · Claude Code · LLM agents · Obsidian  jobhunt-agent	2026
- Event-driven system I run daily: one watcher orchestrates 5 headless LLM workflows over a Markdown knowledge base, from batched job-description scoring and score-gated CV generation to a follow-up outreach queue with a human approval step - The git working tree is the review gate (agents edit, never commit); deterministic \$0 gates and single-session batching keep it roughly 25× cheaper than per-item runs. Open-sourced	
Plant Leaf Disease Classification — Python · TensorFlow · FastAPI · MLflow · pytest	2024
5-model ensemble CNN (transfer learning on MobileNetV2, EfficientNet, ResNet) reaching 86% on unseen test data; a two-stage color-based outlier filter lifted the baseline by 13 points, served behind an async FastAPI API with Grad-CAM explainability and MLflow tracking	
Time-Series Forecasting — Python · TensorFlow · Keras	2024
Benchmarked 6+ sequence architectures (LSTM variants, attention, Transformer) on 48K series with an ACF-driven lookback window; a Conv-LSTM with autoregressive multi-step decoding reached test MSE 0.01	
CNC Machine — 2-DOF Pantograph Robot — MATLAB · Simulink · Quanser SRV02 Hardware	2024
- Modeled, identified, and controlled a real 2-DOF pantograph CNC robot on Quanser hardware: deployed 9 controllers from PID to LQR with a Kalman observer to MPC with input and workspace constraints (best 0.26 s settling, <0.5% overshoot) - Near-perfect 2D trajectory tracking with Kalman filter + LQR, validated in both time and frequency domains on the physical plant	
LANGUAGES	
English (Fluent) Italian (A2) Persian (Native)	